

3.1.5 Enhancement of materials

Content	Potential links to maths and science
Surface decoration – dyeing. Students should be aware of, and be able to describe, dyeing as a surface decoration, including: • vat, discharge and resist, eg tie-dye, batik • when dye is applied in relation to fibres, yarns, fabrics and finished products • the different types of dye fastness required in fashion clothing and textile products, eg fastness to washing, light, perspiration, rubbing and chlorine • dyeing in response to consumer demand and seasonal trends.	Understand the principles behind discharge and resist dyeing. Understand the degradation of fabrics in relation to colour fastness.
Surface decoration – printing. Students should be aware of, and be able to describe, the steps that need to be taken in preparing a fabric for printing, including: • desizing • scouring • bleaching. Students should be aware of, and be able to describe, printing as a surface decoration, including: • direct • discharge • hand (block and stencil) • rotary/flat bed screen printing • transfer • digital printing • dye sublimation.	
Embroidery. Students should be aware of, and be able to describe, a range of hand and machine embroidery stitches.	

Content	Potential links to maths and science
Quilting. Students should be aware of, and be able to describe, the construction of quilted fabrics and reasons for their use.	Understand the effects of trapped air on thermal insulation.